The role of IT leadership in realizing the return on investment of AI coding assistants.

by

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A dissertation submitted in fulfilment of the requirements of the degree of

Master of Commerce

in

Information Technology Management

at the

College of Business and Economics

UNIVERSITY OF JOHANNESBURG

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2026

Abstract

The rapid adoption of generative artificial intelligence (AI) coding assistants is changing how software engineering teams develop, test, document, and deliver software. Despite previous research showing that AI coding assistants can be effective at improving developer productivity, it does not follow that such improvements will result in better ROI. The benefits of integrating AI in a company also depend on how well it manages, promotes and integrates the technology in the routine software development process. This study investigates how organizational culture and strategic leadership influence the realization of ROI from AI coding assistants in software engineering teams. The study is guided by Technology Value Realization Theory as well as the Socio-Technical Systems perspective, which provide a basis for understanding how technological benefits depend on both organizational and human factors.

This research employs the quantitative research design to explore the linkages among various factors like organizational culture, strategic leadership, AI coding assistant assimilation quality, and the return on investments. The study focuses on software engineering teams of medium-scale and large organizations in South Africa that have implemented the AI code assistant technology. It has been planned to collect the data through a well-structured survey with the help of the Likert scale. The return on investment in case of AI coding assistant has been examined on three levels, which are the impact at the level of task, the impacts regarding integration and coordination at the level of workflow, and the effects on the financial and futuristic aspect of the company. AI coding assistant assimilation quality is considered to be the mediating variable along with organizational readiness, the maturity and experience of the team, and length of the adoption as the control variables.

The study aims to fill the gap in existing body of knowledge by examining opportunities to realize profits from the usage of AI programming assistants and the impact of organization-related issues on the overall productivity of programmers. The research will use statistical approaches, e.g., Structural Equation Modeling, to establish the reliability and validity of the measurement model as well as test the relationships between variables in the study. The research is expected to determine if organizational culture and strategic leadership directly impact AI-related ROI and contribute to the proper assimilation of AI coding assistants into software engineering processes.

The study contributes to AI value realization research by linking technology performance to organizational culture and leadership. The study results are expected to provide practical tips for software engineering managers and companies aiming to deploy AI coding assistants in the way that would guarantee positive productivity and business results.

Keywords: Artificial intelligence, AI coding assistants, organizational culture, strategic leadership, technology value realization, assimilation quality, return on investment, software engineering.

Declaration

I, Celumusa H Zulu, certify that the dissertation submitted by me for the Master of Commerce in Information Technology Management degree at the University of Johannesburg is my independent work and has not been submitted by me for a degree at another university.

Celumusa H Zulu

(No signature required)

Dedication

To my Mother, who always believe in me, my kids and to all my Siblings who always believe in me.

To Evelyn Manosa thank you very much for everything you did to make this study possible.

Acknowledgements

First and foremost, infinite gratitude and appreciation to my Creator, Who has given me the will to start studying again and the endurance to see this through in a time when my world was turned on its head. My perseverance, accomplishments and very being are only by Your will.

Also, sincerest gratitude to my Kids, who had to endure my absence whilst completing this study. You are the very foundation my success is built on.

My deepest gratitude to my Mother, my Sister and my brother's as well for always remembering me in your prayers.

My sincerest appreciation to my Supervisor Dr. Siyabonga Mhlongo, who has always been my light through this journey. You got me started on this, thank you for seeing me through to the finish line.

I'd also like to extend my sincerest appreciation to all the people that took their precious time and completed the survey to make this study possible.

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Chapter 1

Introduction

1.1 Introduction

This chapter introduces the study and also provides the background for examining The role of IT leadership in realizing the return on investment of AI coding assistants. It outlines the research problem being addressed, presents the research objectives and research questions of this study. Furthermore the chapter also discusses the significance of the topic and also states the scope and delimitation. Lastly, this chapter will provide the theoretical framework used to guide the study and close off with the dissertation structure.

1.2 Background

Artificial intelligence (AI) is one of the structural factors that has a significant impact on productivity and competitive advantages (Gillespie et al., 2025). As of 2023, generative AI systems have developed from being an experiment to reaching the phase of implementation within organizational processes (Gillespie et al., 2025). Recent reports from around the world show that organizations allocate more money to AI since they expect clear monetary outcomes, cost savings, and improved performance of organizational processes (Gillespie et al., 2025). Research on information systems and management has transcended from the stage of adoption to the understanding of added value meanings with the emphasis on differentiation of organizational value created from investments in AI systems as a function of organizational governance and management performance, not technology excellence (Hossain et al., 2025).

In software engineering, the use of generative AI tools are among the most noticeable forms of AI-enabled augmentation (Peng et al., 2023). Providing developers with functionalities such as code suggestions and entire code functions in real time based on context, documentation, and debugging (Peng et al., 2023). Practical controlled experiments provide conclusive evidence that AI-assisted developers complete software development tasks faster and report increased productivity of about 55.8% compared with control groups (Peng et al., 2023). Collaboration between humans and AI systems is a paradigm shift from the traditional concept of automation, in which AI enhances developers' capabilities rather than replacing them with machines (Peng et al., 2023). Initial performance improvements demonstrated in controlled promising experiments may not necessarily guarantee long-term organizational performance benefits. Empirical studies report considerable variation in the productivity gains from AI-assisted software development tasks across software development teams and organizations.

As it has been long noted, organization culture and management style have been generally believed to affect the implementation success of new technologies. Within the digital field, values and practices will drive the culture of experimentation, learning, and data use in decision making as well as a positive attitude to tech adoption. New findings reveal that digital culture explains the linkage between human AI skills and innovation outcomes, and that culture readiness mediates the performance impact of AI Investments (An et al., 2024). Similarly, this applied also to employees' engagement in AI, ethical AI management or leadership impacts toward strategic alignment. Even though the significance of digital culture and leadership is being gradually recognized by most researchers, current research findings on AI Adoption is more fragmented (An et al., 2024). A significant number of recent works investigated AI adoption under the macro-organizational view or focused on the impacts for personal performance only instead of integrating digital culture, management and ROI in the software industry (Peng et al., 2023).

Generative AI (GenAI) has been increasingly adopted by African companies to address resource constraints and operational inefficiencies. Similarly, there is evidence that developing nations have tried extensively with AI and have positive views about AI's capabilities to boost productivity at work (Gillespie et al., 2025). Yet, literature to date highlights difficulties around digital skills pipelines, governance maturity and leadership capability among others in adopting GenAI technologies (Hughes et al., 2026). In this context, organizational culture and leadership remain particularly critical. If these factors are not supported by a clear strategic vision, trust and a learning orientation, AI adoption is likely to be less about value creation and more about segmentation or symbolism. The digital transformation landscape in Africa is a valid environment for exploring the impact of socio-organizational factors on the achievement of returns from AI technologies.

With regards to software engineering in South Africa, the environment provides a relevant framework in grasping how the various organizations adopt and utilize advanced AI tools. Several studies confirm the fact that AI has emerged as one of the major components that define the transformation of operations on companies and the way they develop their culture due to the increase in number of firms using AI technologies in their business processes (Murire, 2024). There are opportunities and challenges that needs to be addressed in order to leverage organizational performance through AI adoption and alignment with organizational culture and structure, as indicated by research reviews on AI tool adoption in organizations (Romeo & Lacko, 2025). Though the adoption of AI technology affects many aspects of the organization's performance positively, like productivity and output, the effect of AI technology on aspects like the organization's culture and employees morale and engagement is complicated and depends on many factors, as indicated by the study on the impact of AI tools in workplace settings (Kassa & Worku, 2025). Studies on the effect of generative AI tools on software engineering have indicated that AI technology positively affects organizations by enhancing productivity in coding and other software engineering activities through employees adoption of AI tools (Alenezi & Akour, 2025).

Leadership and organizational culture play an important role in the interpretation, adoption and usage of AI technologies by work teams. Research on AI adoption has found that the adoption of AI related technology is also dependent on leadership styles that facilitate embedding technology initiatives into organizational culture (Romeo & Lacko, 2025). Research has also found that the adoption of AI-related technology is dependent on the adoption of the organizational culture that is conducive to the adoption of technology (Kassa & Worku, 2025). Further more research on organizational culture has also shown that cultural elements, particularly the norms that define innovation, collaboration and adaptability, have an impact on the use of AI by staff (Murire, 2024). In the case of software engineering teams that use AI-based coding assistants for various functions, both leadership as well as culture determine the effectiveness of AI in terms of productivity or quality and financial returns.

1.3 Problem statement

Generative AI was primarily used for research and information retrieval purposes in its initial years; however, the use of generative AI has shifted towards the process of knowledge work itself. Experimental results show that generative AI enhances the efficiency and effectiveness of cognitive work (Noy & Zhang, 2023). Furthermore, studies conducted in a work environment setting have also shown improvements in productivity as well as employees performance due to the use of generative AI. It seems from these results that the benefit of generative AI depends upon its implementation rather than the technology itself (Brynjolfsson et al., 2025). More experimental results regarding software development can be found in favour of this theory. Large-scale researches conducted on GitHub Copilot have been showing improvements in the completion time as well as in the efficiency of the development process of developers (Cui et al., 2026). Furthermore, controlled experiments using AI programming assistants have also showed improvements in the speed of completing software development tasks, this provides empirical support for the ROI of generative AI in coding assistance (Peng et al., 2023). All these results collectively show that AI coding assistants are capable of increasing productivity. It is therefore necessary to conduct more research in order to investigate the effects of organizational factors on this phenomenon.

While all these improvements are measurable, research evidence in existing literature has proved that the performance impact varies based on the situation. In software engineering, research has been able to demonstrate that the code generated using GitHub Copilot increases productivity but at the same time creates a level of inconsistency in accuracy, maintainability, and effort required for review purposes; and this clearly demonstrates the need to depend on the management and integration of the system (Moradi Dakhel et al., 2023). Research on the governance of artificial intelligence in organizations has suggested that to achieve the sustainability of value creation, there should be processes and systems in place to ensure accountability and supervision of the leadership to convert experiments into performance improvement (Mäntymäki et al., 2022). Further more studies on organizational innovation with generative AI has demonstrated that the organization will be able to achieve various performance impacts depending on whether the AI is used for automation or augmentation purposes; and this demonstrates the importance of strategic leadership and culture (Holmström & Carroll, 2025). Overall, this existing research evidence shows the role of organizational culture and strategic leadership in determining whether the improvements in productivity due to AI coding assistants can be sustained and converted to measurable return on investment.

The importance of such mechanisms can also be seen within the context of Africa due to the fact that organizations face limitations in terms of skills development, governance, and digitization. The empirical study on the readiness of AI within Africa demonstrated the level of readiness and the organization's ability to implement AI which demonstrates the importance of organizations in the transition from AI pilot to its performance (Isagah & Ben Dhaou, 2024). In the context of South Africa, empirical studies on the financial sector have demonstrated the role of organizational factors, including leadership support and readiness in the adoption of new and emerging technologies which demonstrate the importance of such factors in achieving ROI from using AI coding assistants in software development teams (Matsepe & Van der Lingen, 2022). The empirical research conducted in South African environment highlights the significance of sociotechnical systems to AI adoption and change management as mechanisms of achieving performance outcomes related to AI adoption(Smit et al., 2024). Also the research reveals the significance of managerial perceptions towards AI adoption in achieving performance outcomes within the field of human resources management as the field of AI application.

Collectively, the global evidence shows that AI coding assistants can improve productivity, while organisational research suggests that sustained ROI is influenced by strategic alignment, effective governance, and organisational culture. The African and South African evidence base also indicates that the context is significant in relation to the successful adoption of technology. It is because of this that the evidence base offers a rationale for conducting an in-depth exploration in relation to the role of culture and leadership in achieving ROI with AI coding assistants.

1.4 Research aim and objectives

1.4.1 Main Research Objective

To develop an integrated framework linking financial metrics, organizational culture, and leadership to assess and enhance the return on investment of AI coding assistants in South African software engineering teams.

1.4.2 Sub Research Objectives

1. To identify and operationalize the financial costs and benefits that define return on investment for AI coding assistants.

2. To examine how organizational culture attributes influence productivity and quality outcomes from AI coding assistants.

3. To assess which leadership practices most strongly affect return on investment outcomes.

4. To apply and validate an integrated return on investment framework within large South African organizations.

5. To formulate practical recommendations that enable managers to maximize return on investment through cultural and leadership interventions.

1.5 Research questions

1.5.1 Main Research Question

How do organizational culture and leadership shape the return on investment of AI coding assistants in South African software engineering teams?

1.5.2 Sub Research Questions

RQ1. What financial costs and benefits define return on investment for AI coding assistants in software engineering teams?

RQ2. How do organizational culture attributes influence productivity and quality outcomes from AI coding assistants?

RQ3. Which of the leadership behaviours strongly affect return on investment outcomes from AI coding assistants?

RQ4. How can financial and organizational factors be integrated into a practical return on investment framework for large South African organizations?

RQ5. What practical guidance can be offered to managers to maximize returns from AI coding assistance investments through cultural and leadership interventions?

1.6 Significance/rationale

Generative AI has a potential to enhance productivity in software engineering and efficiency in knowledge work (Brynjolfsson et al., 2025; Peng et al., 2023). AI coding assistants speed up the software development process and make the completion of the task more probable. Nevertheless, even though the productivity has increased, the connection between the use of AI coding assistants and the ROI obtained by software engineering teams remains unclear.

Leadership and organizational culture have been identified as factors that affect the successful adoption of the associated benefits of using AI technology. It affects innovation through guiding organizational resources towards innovative practices (Hossain et al., 2025). A culture of learning, cooperation, and innovation is essential for organizational digital capability (Abdul Wahab & Radmehr, 2024; Li et al., 2025; Poisat et al., 2024). A system perspective, adaptability, is essential for organizational assimilation of digital technologies. Performance of emerging markets is dependent on their level of readiness and leadership context(Gillespie et al., 2025; Nurlia et al., 2023).

However, a link still seems to be missing since there is no literature link between organizational ROI and the productivity of AI coding assistants. Likewise, there is no literature link between organizational ROI and AI value. Finally, the literature on organizational culture and leadership does not have any literature link between organizational ROI and AI coding assistants' value for software engineering teams. This study fills in this gap since it will explore the influence of organizational culture and leadership on ROI of AI coding assistants for software engineering teams.

1.7 Scope and delimitation

1.7.1 Scope

This study is guided by a quantitative research approach that seeks to explore the impact of organizational culture and leadership on the achievement of return on investment (ROI) for AI coding assistants in software engineering teams. The study is limited to the usage of generative AI tools that facilitate the inclusion of code, inline suggestions, debugging, and documentation, as they are part of the software development process. Literature has already proven that the usage of such tools increases productivity as well as the efficiency of task completion (Brynjolfsson et al., 2025; Noy & Zhang, 2023; Peng et al., 2023).

The target population is software engineering teams working for medium-to-large-scale organizations in South Africa who have prior experience working with AI coding assistants for more than six months. This is a reasonable time frame for evaluating the outcomes since the adoption of AI coding assistants is dependent on firm-level characteristics (Babina et al., 2024). Moreover, the digital readiness of the African continent is recognized as an environmental factor (Gillespie et al., 2025).

The scope of independent variables is limited to organizational culture and leadership in this study, as they might impact the achievement of return on investment for AI coding assistants. The organizational culture is analyzed by taking the learning orientation, collaboration, and innovation support as factors (Abdul Wahab & Radmehr, 2024; Fosso Wamba et al., 2024). Leadership is analyzed by taking the strategic direction, governance, and digital support as factors (An et al., 2024; Hossain et al., 2025; Murire, 2024). The return on investment is analyzed by taking the quantified performance indicators as factors.

1.7.2 Delimitation

Because the technical efficiency of AI tools has already been studied in experiments (Peng et al., 2023), it is important to highlight that this research does not focus on the accuracy benchmarking, technical algorithmic evaluation of the efficiency of the AI coding assistants, architecture of the AI model, or other technical evaluations of the efficiency. While in this research, there was a choice to focus on the socio-organizational factors of value realization in the context of AI. The research only includes software engineering teams, not other professional groups using generative AI, such as marketing, finance, or legal teams. This is because the software engineering workflow is significantly different from other professions.

The significance of leadership and organizational culture is considered in the study as the main independent elements of the research. However, some variables which also could have an impact on ROI, including national regulations, macro environment, and pricing models have not been analyzed properly. While it is evident that those variables can considerably influence ROI, the objective of the current research is to focus on only internal variables.

1.7.3 Brief Limitations

There are several constraints that are inherent in this study. To begin with, the calculation of ROI will rely somewhat on self-reported perceptions of productivity and quality. Secondly, the study is limited in terms of violations of generalizability by virtue of its focus on South African organizations. Lastly, the cross-sectional or short-term longitudinal aspect of the design will be without power in terms of the long-term impact on financial results because the effect of the culture and leadership on the productivity might take a long time to manifest itself(An et al., 2024).

1.8 Theoretical and conceptual framework

This study is based on the following premise: while the productivity gains derived from the use of AI coding assistants can be significant for developers, the value that can be derived from the productivity gains for the organization itself remains an open question. This view is consistent with the value realization framework, which argues that business value depends on how the organization manages, supports, and integrates the technology into its day-to-day business processes. It can also connect with the sociotechnical systems perspective noting that, value derived from the utilization of AI coding assistants depends on the people, processes, leaders, and organizations involved.

There is a significant body of research that suggests that generative AI has a positive impact on individual performance. Research conducted in the field has shown that generative AI has a positive effect on the productivity and quality of performance (Brynjolfsson et al., 2025). Experimental research has also shown that generative AI has a positive effect on the efficiency and quality of performance (Noy & Zhang, 2023). Research has shown that programmers working with AI assistants perform their tasks more efficiently and with better quality than those programmers who do not use AI assistants (Peng et al., 2023). However, research conducted at the firm level has shown that the ROI generated by AI is firm-specific, depending upon how the firm invests in AI (Babina et al., 2024).

In this context, leadership and organizational culture are identified as key factors that explain the disparities identified. Leadership is responsible for guiding and directing the firm, as well as the way AI coding assistants are used within the firm. The empirical evidence suggests that leadership is a key factor that affects digital or AI performance outcomes (Hossain et al., 2025; Murire, 2024). On the other hand, organizational culture is responsible for influencing the behavioural aspects of individuals within the firm. Learning culture is a culture that promotes the use of AI tools within the firm through collaboration and innovation. The empirical evidence suggests that a digital organizational culture strengthens the relationship between AI performance outcomes and organizational performance (Fosso Wamba et al., 2024). Furthermore, a digital organizational culture increases innovation outcomes by human-AI capabilities (An et al., 2024).

The conceptual model in question defines ROI from AI coding assistants as the key variable of particular interest are the roles of IT leadership and organizational culture in relation to ROI as the explaining variables in the model. In addition, the importance of assimilation quality is explained in the context of the mediating role of leadership and organizational culture in their effect on ROI. Assimilation quality is understood as the stage in which AI coding assistants are integrated into the process of software engineering and start being used in existing workflows. Furthermore, organizational readiness is included in the model as a control variable, the potential for successful implementation of AI coding assistants would depend on the level of organizational readiness to adopt and work with digital technologies in general. Even though there is available literature reporting on higher performance on the level of individuals and tasks, little is known about the relationship between the achieved productivity level and the ROI from AI coding assistants on the level of organizations (Gillespie et al., 2025; Nurlia et al., 2023).

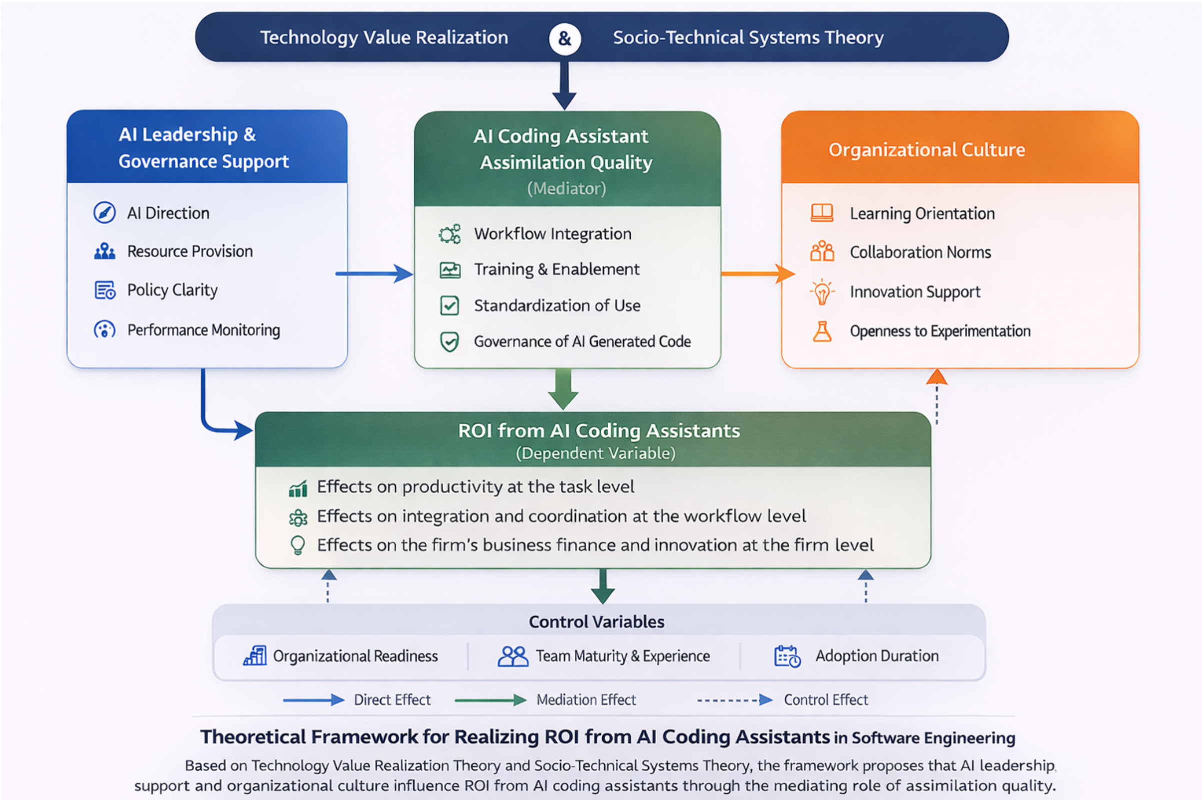

Figure 1.1: theoretical framework.

1.9 Dissertation structure

This dissertation has about 6 chapters outlined below and in Figure 1.2:

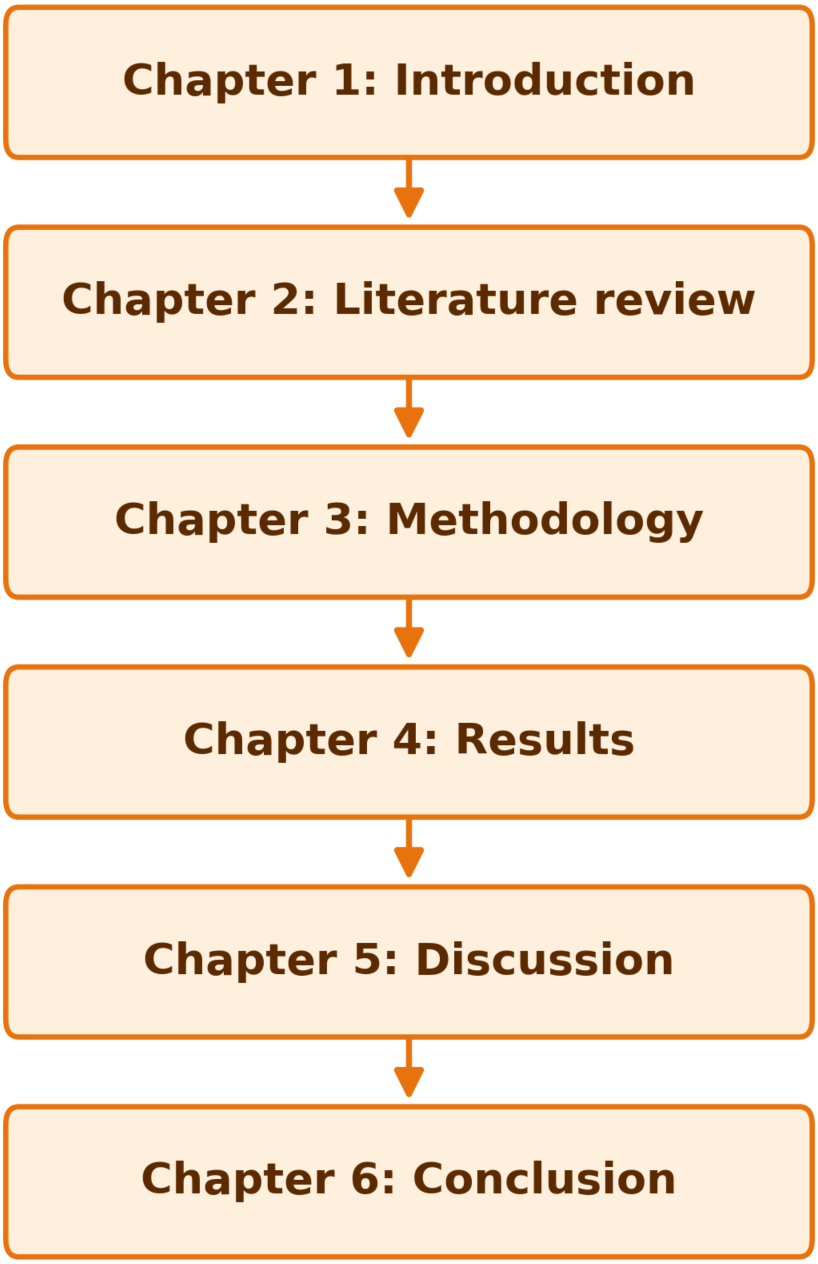

Figure 1.2: dissertation structure.

Chapter 1: Introduction

Chapter 1 introduces the study by providing the background for examining The role of IT leadership in realizing the return on investment of AI coding assistants. It outlines the research problem being addressed, presents the studies objectives and research questions of this study. Furthermore the chapter also discusses the significance of the topic and also states the scope and delimitation. Lastly, this chapter provides the theoretical framework used to guide the study and closed off with the dissertation structure.

Chapter 2: Literature review

Chapter 3: Methodology

Chapter 4: Results

Chapter 5: Discussion

Chapter 6: Conclusion

1.10 Chapter summary

Chapter 2

Literature Review

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Chapter 3

Research Methodology

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Chapter 4

Results

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Chapter 5

Discussion

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Chapter 6

Conclusion

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Appendix A

Ethical Clearance Certificate

UNIVERSITY OF JOHANNESBURG

CBE RESEARCH ETHICS COMMITTEE

Dear Researchers

ETHICAL APPROVAL GRANTED FOR RESEARCH PROJECT

Decision: Clearance granted

This letter serves to confirm that that the proposed research project indicated in the table below, has been reviewed by the Department of Applied Information Systems at the University of Johannesburg. Ethical clearance is hereby granted and is valid for three years, from 01 January 1900 until 31 December 2999.

Applicant	
Supervisor	
Student/staff number	
Title	
Decision date at meeting	
Reviewers	
Ethical clearance code	
Rating of application	

CODE 01 –	Approved	CODE 02 –	Approved with suggestions/requirements with no re-submission
CODE 03 –	Referred back	CODE 04 –	Disapproved, cannot re-submit

The researcher/s may now commence with the study providing that:

1. The researcher/s will ensure that the project adheres to ethical research requirements
2. The researcher/s will be conducting the study as set out in the approval application
3. The researcher/s will ensure that project adheres to all applicable legislation, scopes of practice, professional codes of conduct and scientific standards as it pertains to the field or study.
4. The researcher/s will bring under the attention of the research ethics committee any proposed changes, concerns that arise, and unexpected ethical management issues
5. Any changes that can affect the study-related risks for participants or researchers must be reported to the committee in writing
6. Only data collected during the three valid period is permissible and valid for the study

7. No data collect before or after the three valid period is ethically valid and the Research Ethics Committee takes no responsibility for any issues arising in this situation
8. No fieldwork activities may continue after the ethical clearance has expired. A request for an extension of ethical clearance can be made in writing to the REC
9. All data collection must be done within the Government regulations of the lockdown
10. The researcher/s must be aware that COVID-19 may influence the responses and needs to be controlled for

Furthermore, the researcher/s must please address the following comments from the ethics committee:

- N/A

Prof M Name

AIS REC

aisrec@uj.ac.za

Appendix B

Appendix Chapter

B.1 Appendix section

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Appendix C

Another Appendix Chapter

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